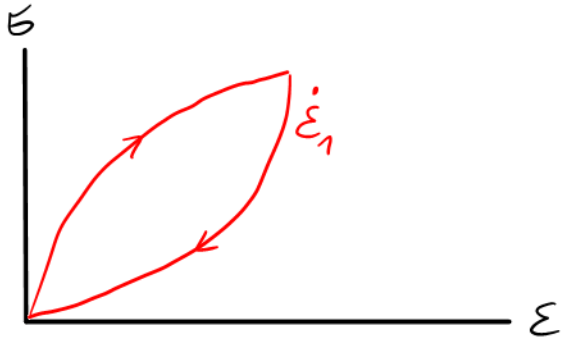
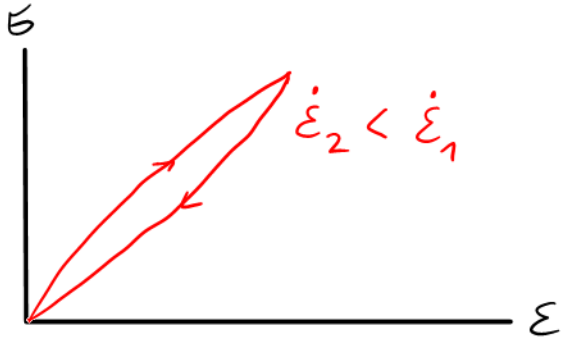
Visco-elasticity

Behavior of visco-elastic materials

Cyclic response



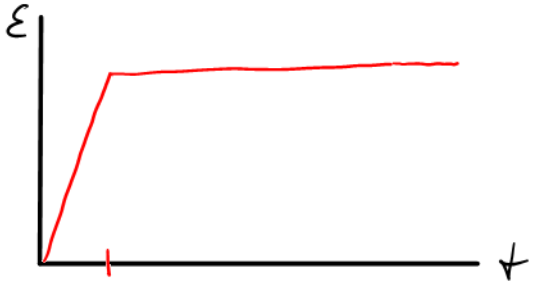


⇝ stress-strain hysteresis

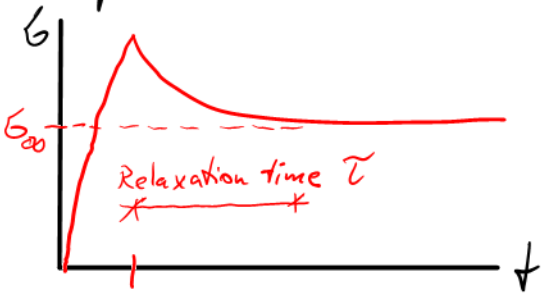
⇝ rate dependency

Relaxation

Protocol:

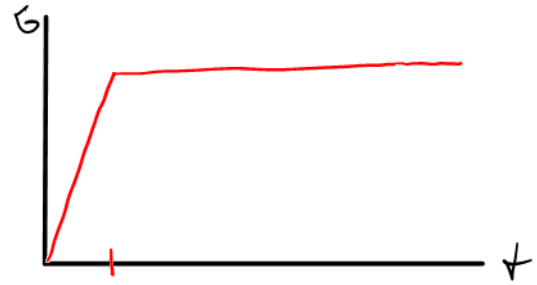


Response:



Creep

Protocol:

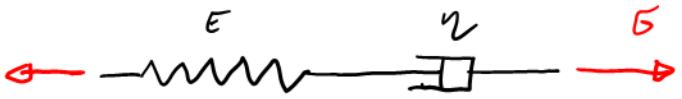


Response:



Rheological models

Maxwell model (spring and damper acting as series)



- Strain: $\varepsilon = \varepsilon_E + \varepsilon_\eta$
- Stress: $\sigma = \sigma_E = \sigma_\eta$ (equilibrium of forces)
- Material laws: $\varepsilon_E = \frac{1}{E}\sigma$, $\dot{\varepsilon}_\eta = \frac{1}{\eta}\sigma$

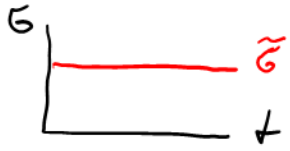
$$\rightsquigarrow \dot{\varepsilon} = \dot{\varepsilon}_E + \dot{\varepsilon}_\eta = \frac{1}{E}\dot{\sigma} + \frac{1}{\eta}\sigma \quad | \cdot E$$

$$\Leftrightarrow E\dot{\varepsilon} = \dot{\sigma} + \frac{1}{\tau}\sigma \quad \text{with relaxation time } \tau = \frac{\eta}{E} \qquad (1)$$

ordinary differential equation of 1st order → evolution eq.

- Creep response: (due to stress jump)

$$\sigma(t) = \tilde{\sigma}$$



$$\rightsquigarrow \dot{\sigma} = 0 \quad \text{für } t > 0$$

$$\rightsquigarrow \text{ODE}^{(1)}: \quad E\dot{\varepsilon} = \cancel{\dot{\sigma}}^{=0} + \frac{1}{\tau}\tilde{\sigma} \quad \Leftrightarrow \quad \dot{\varepsilon} = \frac{1}{\tau E}\tilde{\sigma} = \frac{\tilde{\sigma}}{\eta}$$

$$\Leftrightarrow \int_t \dot{\varepsilon}\, dt = \int_t \frac{\tilde{\sigma}}{\eta}\, dt$$

$$\Leftrightarrow \varepsilon(t) = \frac{\tilde{\sigma}}{\eta}t + c$$

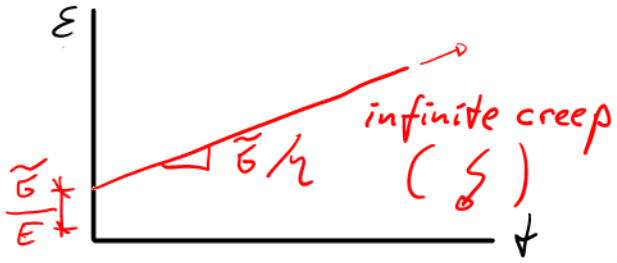
Integration constant from initial condition

$$\varepsilon(0) = \varepsilon_E(0) + \varepsilon_\eta(0)$$

$$= \frac{\tilde{\sigma}}{E} + 0 \quad \rightsquigarrow \quad \frac{\tilde{\sigma}}{E} = C$$

Solution of ODE

$$\varepsilon(t) = \tilde{\sigma}\underbrace{\left(\frac{1}{\eta}t + \frac{1}{E}\right)}_{\text{creep function}}$$



- Relaxation response: (due to strain jump)

$$\varepsilon(t) = \tilde{\varepsilon} \quad \rightsquigarrow \quad \dot{\varepsilon} = 0 \text{ for } t > 0$$

(1): for $\dot{\varepsilon} = 0 \Rightarrow \underbrace{\dot{\sigma} + \frac{1}{\tau}\sigma = 0}_{\text{homogeneous ODE of 1st order}}$

$\rightsquigarrow$ evolution equation

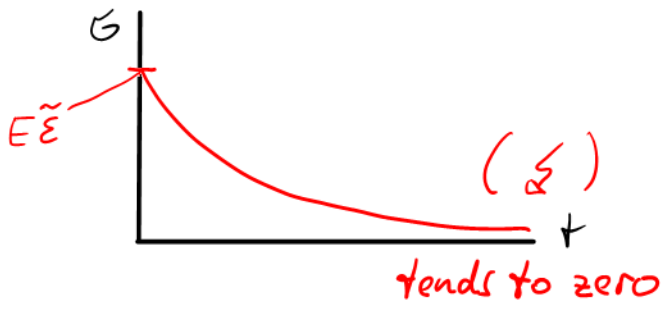
Solution of ODE:

$$\sigma(t) = A\, e^{-\frac{1}{\tau}t}$$

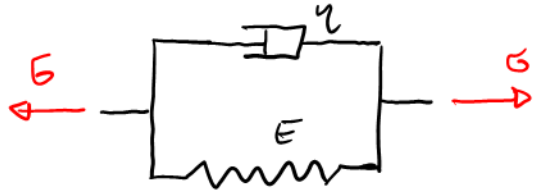
$$AB: \sigma(0) = E\tilde{\varepsilon}$$

$$\rightsquigarrow E\tilde{\varepsilon} = A e^{-\frac{1}{\tau}\cdot 0} = A$$

$$\Rightarrow \sigma(t) = \tilde{\varepsilon}\underbrace{E e^{-\frac{t}{\tau}}}_{\text{Relaxation function}}$$



Kelvin-Voigt model (spring and damper acting in parallel)



- Strains: $\varepsilon = \varepsilon_E = \varepsilon_\eta$
- Stresses: $\sigma = \sigma_E + \sigma_\eta$

$$= E\varepsilon + \eta\dot{\varepsilon}$$

$$= E(\varepsilon + \tau\dot{\varepsilon})$$

- Creep response (stress jump)

$\sigma(t) = \tilde{\sigma}\, H(t)$, Heaviside function $H(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t \geq 0 \end{cases}$

for $t \geq 0 \rightsquigarrow H = 1 \rightsquigarrow \sigma = \tilde{\sigma}$

ODE: $\tilde{\sigma} = E\varepsilon + E\tau\dot{\varepsilon} \quad | \cdot \frac{1}{E\tau}$

$$\Leftrightarrow \underbrace{\frac{\tilde{\sigma}}{E\tau}}_{\text{const.}} = \dot{\varepsilon} + \frac{1}{\tau}\varepsilon \qquad \text{(non-homogeneous ODE of 1st. o.)}$$

Solution:

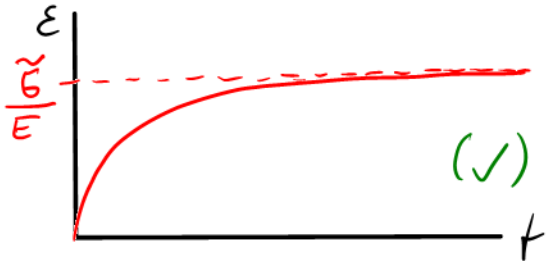
$$\left.\begin{aligned} \varepsilon_{part} &= \frac{\tilde{\sigma}}{E} \\ \varepsilon_{hom} &= c\, e^{-\frac{t}{\tau}} \end{aligned}\right\} \quad \varepsilon(t) = \frac{\tilde{\sigma}}{E} + c\, e^{-\frac{t}{\tau}}$$

Initial condition: $\varepsilon(0) = 0$

$$\rightsquigarrow \frac{\tilde{\sigma}}{E} + c \cdot 1 = 0$$

$$\Leftrightarrow \quad c = -\frac{\tilde{\sigma}}{E}$$

$$\Rightarrow \varepsilon(t) = \tilde{\sigma}\, \underbrace{\frac{1}{E}\left(1 - e^{-\frac{t}{\tau}}\right)}_{\text{creep function}}$$



- Relaxation response (strain jump)

$$\varepsilon(t) = \tilde{\varepsilon}\, H(t)$$

$$\rightsquigarrow \dot{\varepsilon}(t) = \tilde{\varepsilon}\, \delta(t), \quad \text{Dirac-function } \delta(t) = \begin{cases} \infty, & t=0 \\ 0, & t\neq 0 \end{cases}$$

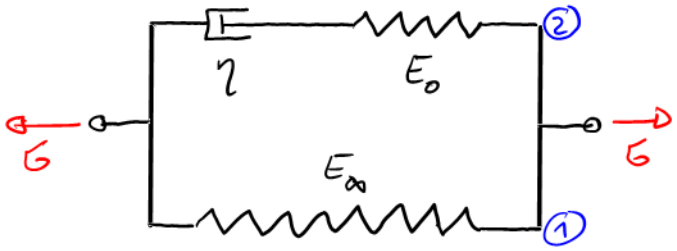
$$\rightsquigarrow \sigma(t) = E\,\varepsilon(t) + \eta\,\dot{\varepsilon}(t)$$

$$= \tilde{\varepsilon}\, E\left(H(t) + \underbrace{\frac{\eta}{E}}_{\tau}\,\delta(t)\right) \quad \rightsquigarrow \sigma(t=0) \to \infty$$

relaxation function

(↯)

3 Parameter model



$$\sigma = \sigma_{①} + \sigma_{②}$$

$$\varepsilon_{①} = \varepsilon_{②} = \varepsilon$$

with $\sigma_{①} = E_\infty \varepsilon$

$$\sigma_{②} = \eta \dot{\alpha} = E_0 (\varepsilon - \alpha) \qquad (2)$$

$\varepsilon - \alpha = \varepsilon^{el}$ ≙ strain in spring

α ≙ strain in damper

$$\rightsquigarrow \sigma = E_\infty \varepsilon + E_0 (\varepsilon - \alpha) \qquad (3)$$

$E_0(\varepsilon - \alpha)$ = overstress

Evolution equation for α:

$$(2): \quad \dot{\alpha} = \frac{E_0}{\eta} (\varepsilon - \alpha) = \frac{1}{\tau} (\varepsilon - \alpha)$$

relaxation time $\tau = \frac{\eta}{E_0}$

$\rightsquigarrow$ corresponds to additive split in element ②

$$\varepsilon^{el} = \varepsilon - \alpha \quad \text{and} \quad \varepsilon^{in} = \alpha$$

ε^{in}: inelastic

Integration of evolution equation:

$$\dot{\alpha} + \frac{1}{\tau} \alpha = \frac{1}{\tau} \varepsilon \qquad | \cdot e^{t/\tau}$$

ε: not const. (4)

$$\Leftrightarrow \underbrace{\left(\dot{\alpha} + \frac{1}{\tau} \alpha \right) e^{t/\tau}}_{\hat{=} \frac{d}{dt} (\alpha e^{t/\tau})} = \frac{1}{\tau} \varepsilon e^{t/\tau}$$

$$\Leftrightarrow \frac{d}{dt} \left(\alpha e^{t/\tau} \right) = \frac{1}{\tau} \varepsilon e^{t/\tau}$$

Solution:

$$\alpha = \frac{1}{\tau}\int_{-\infty}^{t} \underbrace{e^{-\frac{t-s}{\tau}}}_{\dot{u}} \underbrace{\varepsilon(s)}_{v}\, ds \quad , \qquad \alpha(t \to -\infty) = 0$$

$$u = \tau e^{-\frac{t-s}{\tau}}$$

partial integration:

$$\rightsquigarrow \int_{-\infty}^{t} \dot{u}\, v\, ds = [u\,v]\Big|_{-\infty}^{t} - \int_{-\infty}^{t} u\, \dot{v}\, ds$$

$$= \left[\tau e^{-\frac{t-s}{\tau}} \varepsilon(s)\right]\Big|_{-\infty}^{t} - \int_{-\infty}^{t} \tau e^{-\frac{t-s}{\tau}} \dot{\varepsilon}(s)\, ds$$

$$= \tau\,\varepsilon(t) - \int_{-\infty}^{t} \tau e^{-\frac{t-s}{\tau}} \dot{\varepsilon}(s)\, ds$$

$$\Rightarrow \alpha = \varepsilon - \int_{-\infty}^{t} e^{-\frac{t-s}{\tau}} \dot{\varepsilon}(s)\, ds \qquad (5)$$

insert in constitutive equation:

$$\sigma \overset{(3)}{=} E_\infty \varepsilon + E_0(\varepsilon - \alpha)$$

$$= E_\infty \varepsilon + E_0 \int_{-\infty}^{t} e^{-\frac{t-s}{\tau}} \dot{\varepsilon}(s)\, ds$$

$$= \int_{-\infty}^{t} \left(E_\infty + E_0 e^{-\frac{t-s}{\tau}}\right) \dot{\varepsilon}(s)\, ds$$

Thermodynamic formulation

Thermodynamic model for linear 1D case

here: internal variable is strain-like (see above)

- Energy: $\Psi := \Psi(\varepsilon, \alpha)$

$$= \frac{1}{2} E_\infty \varepsilon^2 + \frac{1}{2} E_0 (\varepsilon - \alpha)^2$$

$$\rightsquigarrow \dot{\Psi} = \frac{\partial \Psi}{\partial \varepsilon} \dot{\varepsilon} + \frac{\partial \Psi}{\partial \alpha} \dot{\alpha}$$

$$= \left(E_\infty \varepsilon + E_0 (\varepsilon - \alpha) \right) \dot{\varepsilon} - E_0 (\varepsilon - \alpha) \dot{\alpha}$$

Insert in Clausius-Duhem inequality:

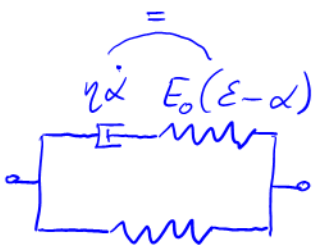
$$\sigma \dot{\varepsilon} - \dot{\Psi} = \sigma \dot{\varepsilon} - \left(E_\infty \varepsilon + E_0 (\varepsilon - \alpha) \right) \dot{\varepsilon} + E_0 (\varepsilon - \alpha) \dot{\alpha} \geq 0$$

- $\sigma = E_\infty \varepsilon + E_0 \underbrace{(\varepsilon - \alpha)}$

non equilibrial state since $\alpha = \alpha(\varepsilon, t)$

- $\underbrace{E_0 (\varepsilon - \alpha) \dot{\alpha}}_{= \sigma^{in} = \eta \dot{\alpha}} \geq 0$

$$\rightsquigarrow \eta \dot{\alpha}^2 \geq 0 \quad \checkmark$$



- Numerical calculation of α :

Introduce time discretization $\Delta t = t_{n+1} - t_n$ and

$$(5): \quad \varepsilon(t) - \alpha(t) = \int_{-\infty}^{t} e^{-\frac{t-s}{\tau}} \dot{\varepsilon}(s)\,ds$$

$t \rightsquigarrow t_{n+1}$:

$$\varepsilon_{n+1} - \alpha_{n+1} = \int_{-\infty}^{t_{n+1}} e^{-\frac{t_{n+1}-s}{\tau}} \dot{\varepsilon}(s)\,ds$$

$$= \int_{-\infty}^{t_n} e^{-\frac{t_n+\Delta t-s}{\tau}} \dot{\varepsilon}(s)\,ds + \int_{t_n}^{t_{n+1}} e^{-\frac{t_{n+1}-s}{\tau}} \dot{\varepsilon}(s)\,ds$$

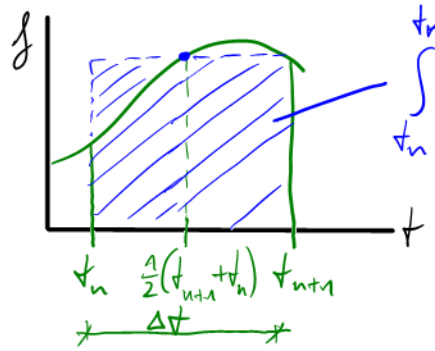
$$\text{NR}: \; e^{-\frac{t_n+\Delta t-s}{\tau}} = e^{-\frac{t_n-s}{\tau}} \cdot e^{-\frac{\Delta t}{\tau}}$$

independent from s

$$= e^{-\frac{\Delta t}{\tau}} \int_{-\infty}^{t_n} e^{-\frac{t_n-s}{\tau}} \dot{\varepsilon}(s)\,ds + \int_{t_n}^{t_{n+1}} e^{-\frac{t_{n+1}-s}{\tau}} \dot{\varepsilon}(s)\,ds$$

$\hat{=} \varepsilon_n - \alpha_n$ (I)

(I) by midpoint rule



$$\int_{t_n}^{t_{n+1}} f(t)\,dt \approx f\left(\tfrac{1}{2}(t_{n+1}+t_n)\right)\Delta t$$

$$\textcircled{I} \approx e^{-\frac{\sigma_{n+1} - \frac{\sigma_{n+1}}{2} - \frac{\sigma_n}{2}}{\tau}} \underbrace{\dot{\varepsilon}\left(\frac{\sigma_{n+1}+\sigma_n}{2}\right)\Delta t}$$

const. rate in Δt:

$$\dot{\varepsilon} = \frac{\varepsilon_{n+1} - \varepsilon_n}{\Delta t} \rightsquigarrow \dot{\varepsilon}\,\Delta t \approx \varepsilon_{n+1} - \varepsilon_n =: \Delta\varepsilon$$

$$= e^{-\frac{\Delta t}{2\tau}} \Delta\varepsilon$$

insert $\textcircled{I}$

$$\varepsilon_{n+1} - \alpha_{n+1} = e^{-\frac{\Delta t}{\tau}}(\varepsilon_n - \alpha_n) + e^{-\frac{\Delta t}{2\tau}}(\varepsilon_{n+1} - \varepsilon_n)$$

$$\Leftrightarrow \alpha_{n+1} = \varepsilon_{n+1} - e^{-\frac{\Delta t}{2\tau}}\varepsilon_{n+1} + e^{-\frac{\Delta t}{2\tau}}\varepsilon_n - e^{-\frac{\Delta t}{\tau}}\varepsilon_n + e^{-\frac{\Delta t}{\tau}}\alpha_n$$

$$= \left(1 - e^{-\frac{\Delta t}{2\tau}}\right)\varepsilon_{n+1} + \left(e^{-\frac{\Delta t}{2\tau}} - e^{-\frac{\Delta t}{\tau}}\right)\varepsilon_n + e^{-\frac{\Delta t}{\tau}}\alpha_n$$

history variables

$$= \alpha_{n+1}(\varepsilon_{n+1}, \sigma_{n+1})$$

Thermodynamic formulation in 3D

here: internal variable is stress-like
→ direct consideration of overstress as internal variable

- Stress:

$$\underline{\sigma} = \underbrace{\frac{\partial \psi^{\infty}}{\partial \underline{\varepsilon}}}_{\underline{\sigma}^{\infty}} - \underbrace{\underline{\alpha}}_{\underline{\alpha} = \underline{\alpha}(\underline{\varepsilon}, t)}$$

- Evolution equation analogous to (4), but stress-related:

$$\dot{\underline{\alpha}} + \frac{1}{\tau}\underline{\alpha} = \frac{\beta}{\tau}\underline{\sigma}^{\infty}$$

(calculation of α by numerical integration)